

Lecturer's name:	Date:
Course: 6th Grade Math	Duration: 1 class period
Unit: 02: Ordering Fractions, Decimals, and Integers	TEKS: 6.2(D) order a set of rational numbers arising from mathematical and real-world contexts
Topic: Things are Not Always as They Seem	
Lesson objectives: Students will be able to discern the true cost of items based on knowledge of ordering rational numbers.	
Essential Questions: Can numbers lie? Justify your answer.	
LYS Methodology Be (Character/Values/Principles) - Thriftiness Know (What resources are available to the student/ How can they use them) - Money Management Do (Execute with Excellence) - Stay on Budget	
Resources (Include Technology): Student: computer, notebook and paper Classroom: Internet, computer, and projector	
Instructional Input AS: Anticipatory Set (Introduction) As students come into the classroom, have the Essential Question on the board and expect students to begin answering it in their notebooks (live classroom) or in a shared google docs document (virtual). As students answer the question, take note of common answers. Select one student representing each answer to share with the group. After each selected student shares their answer, allow students with different viewpoints to share why they disagree. After all students who want to share have had an opportunity, transition to the lesson by saying, "Today we are going to look at some numbers that do not look like what they are up front. Then we will return to this question. I'm excited to see if you have changed your mind by the end of the lesson." M: Modeling (I do) Show the Khan Academy video: https://youtu.be/VZOHWaw5dqM GP: Guided Practice (We do): <i>Now let's look at how ordering rational numbers can make us better decision makers.</i> Google Slides: https://drive.google.com/file/d/1qtudQZ2rCIIZnLHKQz59i8mJ3aJAfa_v/view?usp=sharing Show the table in the google slides and ask: Slide 1:	

- *What information does this table give us?*
- *At first glance, which car seems the most affordable to purchase tires for?*
- *What do we need to do to figure it out?*
 - Have all students write down their answer before continuing.

Slide 2:

- Students independently turn 200/60,000 to 1/300; then check work and give the correct answer; make sure students know this number is \$x per y miles
- Students independently turn 100/25,000 to 1/250; then check work and give the correct answer

Slide 3:

- Students independently turn 150/40,000 to 0.00375
- Students independently turn 150/25,000 to 0.006
- *Now let's take our four numbers and order them.*
- Have students turn all numbers into decimals and order independently answer the question, *Which is the most affordable car to purchase based on tire cost?*; post the independent activity (slide 4) on board and students who have it answered correctly can move onto this independent project.
- **Answer: Economy, Full Size, Midsize, Truck**

Independent Practice/ Homework

- Go to <https://www.goodyear.com/en-US/tires/auto-tires>
- Select "Vehicle Types"
 - Type here the type of vehicle you selected: _____
- Pick four tires. Click on the "view details" of each tire. The tires you select **must** have a "Tread Life Warranty." This means Goodyear expects the tires to last that many miles. Fill in the table below with the four tires you picked:

Name of Tire	Cost of Tire	Tread Life Warranty

- Which tire is the most affordable? How do you know? Show your work below.

Lesson Close

When all students have the correct answer, pull students back together and ask,

- *Was your prediction correct?*
- *Let's return to our original question. Can numbers lie? Do you still think the same as before this lesson? Why or why not?*
 - Allow students to reflect and discuss answers.